

Digital Signal Processing ENGR 4333/5333

Sample Test 1

Date:

Time: 60 minutes

Name:

Q1) Consider a system described by the impulse response $h(t) = t e^{-20t} \cos(28\pi t) u(t)$. Assuming that $h(t)$ is sampled at a rate of 50 Hz, find the following:

- a) The sampled signal $h[n]$.
- b) What is the fundamental period N_0 .
- c) Is the system stable? Give the reason to your answer.

Q2) For the signal $x[n] = n (u[n] - u[n-5])$

- a) Plot $x[2n]$
- b) Find the energy of the signal $x[n]$.

Q3) Determine the zero input response for the system specified by the difference equation below and initial conditions $y[-1] = 1$ and $y[0] = 2$

$$y[n+2] - y[n+1] + 0.16y[n] = 4x[n+2]$$

Q4) Determine a closed-form representation of the unit impulse response $h[n]$ for the system specified by the difference equation

$$y[n] - y[n-1] + 0.25y[n-2] = 4x[n]$$